



**EURO**

# Fundamental Quantum Computing Algorithms and Their Implementation in Qiskit

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IT4Innovations  
VSB - Technical University of Ostrava

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IT4INNOVATIONS  
NATIONAL SUPERCOMPUTING  
CENTER

# Aims of the training

- ❶ Familiarize you with the quantum computing
- ❷ Show the possible advantage of quantum computers on specific tasks

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# Table of Contents



## 1 Day 1

- Introduction to Quantum Computers and Quantum Computing
- Quantum entanglement
- Quantum teleportation
- Bernstein-Vazirani algorithm
- Simon's algorithm
- Q&A and Closing the day

# Timetable of day 1

- 9:00-10:30 Introduction to Quantum Computers and Quantum Computing
- 10:30-10:45 Break
- 10:45-12:00 Quantum entanglement
- 12:00-13:00 Lunch Break
- 13:00-14:00 Quantum teleportation
- 14:00-15:00 Bernstein-Vazirani algorithm
- 15:00-15:15 Break
- 15:15-16:15 Simon's algorithm
- 16:15 Q&A and Closing the day

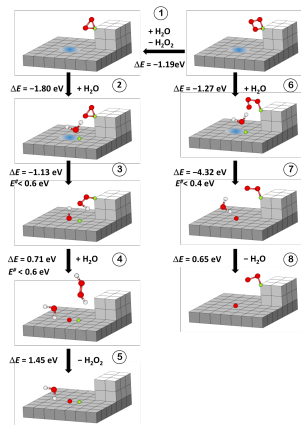
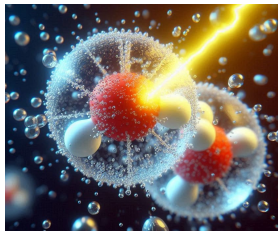
**Questions welcome!!!**

University of Chemistry and Technology Prague  
(2014-2025):

- Physical chemistry
- Computational chemistry

VŠB-Technical University Ostrava (2024 - present):

- Quantum Computing Lab
- 



- Institute of VSB – Technical University of Ostrava
- Supercomputing and research center
- The most powerful supercomputing systems in the Czech Republic.
- Five research labs
- High-Performance Computing, Data Analysis, Quantum Computing, and Artificial Intelligence



# First task

Please, introduce yourself.

Preferably in chat:

Something like name, institute, field of main expertise and knowledge of quantum mechanics

Questions and even comments welcome at any time and in any way!

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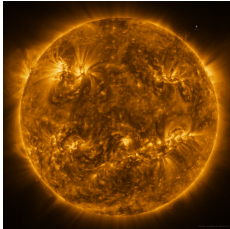
## 1 Day 1

- Introduction to Quantum Computers and Quantum Computing
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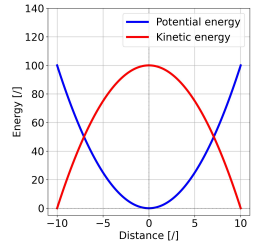
# Classical vs Quantum Mechanics

## Classical Mechanics

Sun



Chevrolet Impala 1967



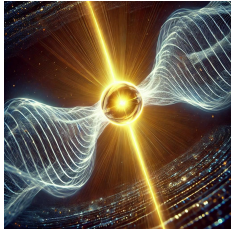
What would happen with the decreasing size?

- Weight:  $\text{kg} \rightarrow 10^{-31} \times \text{kg}$
- Size:  $\text{m} \rightarrow 10^{-10} \times \text{m}$

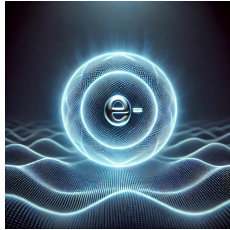
# Classical vs Quantum Mechanics

## Quantum Mechanics

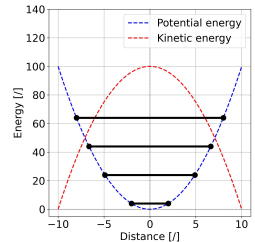
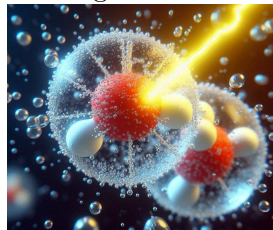
Photon



Electron



Light atoms

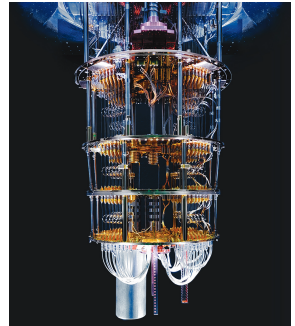
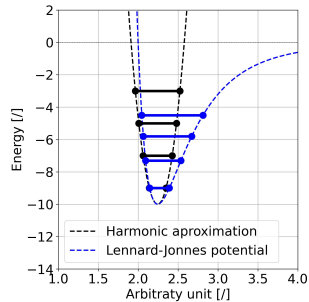
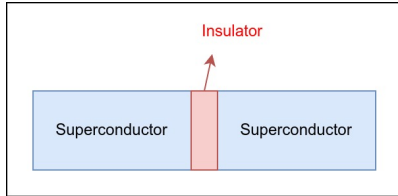


- Stern-Gerlach experiment
- Black-body radiation
- Zeeman effect

# Quantum Hardware

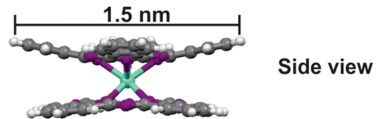
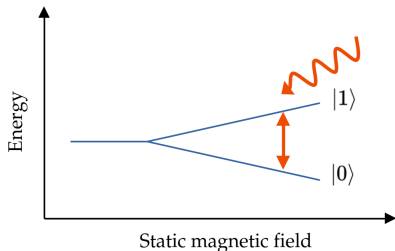
Basic idea: To facilitate the discrete energetic levels

Superconducting technology - Josephson junction

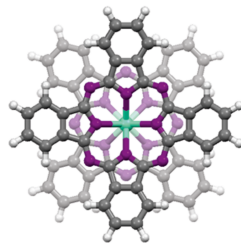


## QC based on molecular spin

- Single-ion Magnets
- f-block ion in the middle – 1 “unpaired electron”
- Highly adjustable ligands



Side view



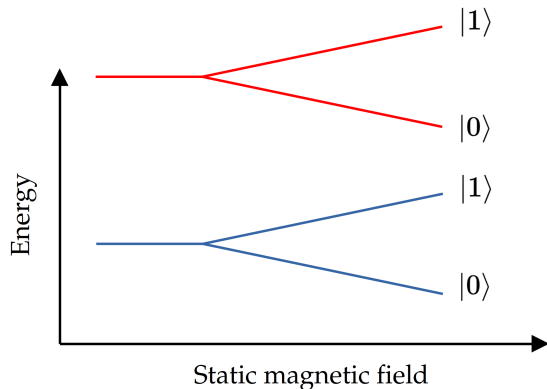
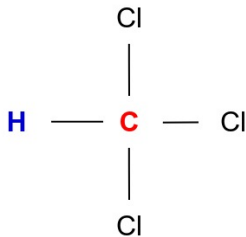
Top view

○ Hydrogen  
● Carbon  
● Nitrogen  
● Terbium

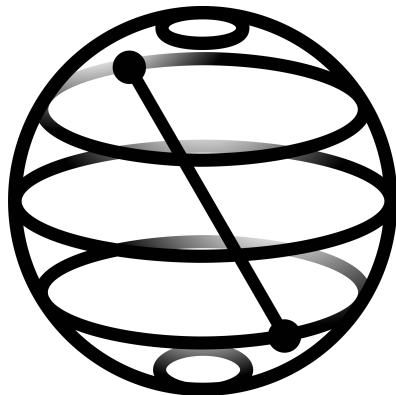
<https://pubs.acs.org/doi/pdf/10.1021/acs.jpcc.6b03676>

## QC based on NMR

- Atoms with odd number of protons and/or neutrons.
- $^1\text{H}$ ,  $^{13}\text{C}$ ...  $^{19}\text{F}$ ,  $^{31}\text{P}$



- Qiskit = Quantum Information Software Kit
- open-source software development kit (SDK) for working with quantum Computers



**Quantum bit (qubit)** is a two-dimensional quantum mechanical system that is in the state:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad (1)$$

where  $\alpha$  and  $\beta$  are complex numbers (the amplitudes of the quantum states  $|0\rangle$  and  $|1\rangle$ , respectively), and they satisfy the normalization condition:

$$|\alpha|^2 + |\beta|^2 = 1 \quad (2)$$

This definition uses the **standard bra-ket notation**, where quantum states are represented as vectors:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (3)$$

# Quantum Registers

- Each bit configuration in quantum superposition is represented as the tensor product of individual qubits.
- For example, the state  $|10\rangle$ , which represents the number 2 in a bit string, is written as:

$$|10\rangle = |1\rangle \otimes |0\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad (4)$$

# Bloch sphere

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

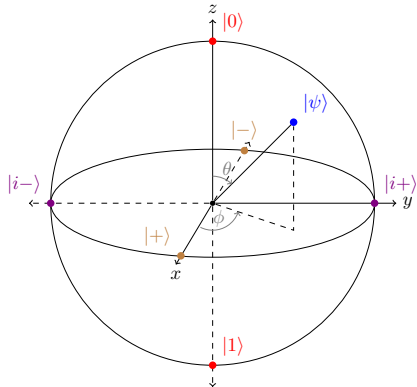
$$\alpha = \cos(\theta/2)$$

$$\beta = e^{i\phi} \sin(\theta/2)$$

$$P(|0\rangle) = |\alpha|^2 = \cos^2(\theta/2)$$

$$P(|1\rangle) = |\beta|^2 = |e^{i\phi} \sin(\theta/2)|^2 = \sin^2(\theta/2)$$

$$P(|0\rangle) + P(|1\rangle) = \cos^2(\theta/2) + \sin^2(\theta/2) = 1$$



# Bloch sphere

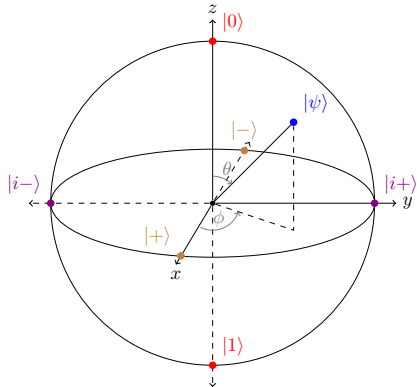
Other interesting point on Bloch sphere:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$|i+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$$

$$|i-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$$



# One-Qubit Quantum Gates

**Definition:** A one-qubit gate is represented by a  $2 \times 2$  unitary matrix  $U$  acting on a single qubit:

$$U|\psi\rangle = U \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

**Important One-Qubit Gates:**

- **Pauli Gates:**

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

- **Hadamard Gate:**

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

- **Phase Gates:**

$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}, \quad T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$$

# Actions of One-Qubit Quantum Gates

## Example:

- Pauli-X (NOT gate):  $X |0\rangle = |1\rangle$ ,  $X |1\rangle = |0\rangle$
- Hadamard on  $|0\rangle$ :

$$H |0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \left( \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

- Hadamard on  $|1\rangle$ :

$$H |1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

- Z-gate on superposition state:

$$Z \left( \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \right) = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

# Actions of One-Qubit Quantum Gates II

**Initial State:** The superposition state

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

**Action of the  $S$  Gate:**

$$S|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle) = |i+\rangle$$

**Quantum Circuit:**

$$|0\rangle \text{ --- } \boxed{H} \text{ --- } \boxed{S} \text{ --- } |i+\rangle$$

The  $S$  gate introduces a phase factor  $i$  to  $|1\rangle$ , transforming  $|+\rangle$  into  $|i+\rangle$ , an eigenstate of the  $Y$  operator.

$$P(\lambda) |\psi\rangle = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\lambda} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ e^{i\lambda} \beta \end{bmatrix}$$

$$Z |\psi\rangle = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ -\beta \end{bmatrix}$$

$$S |\psi\rangle = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{2}} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ i\beta \end{bmatrix}$$

$$T |\psi\rangle = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{4}} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ e^{i\frac{\pi}{4}} \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ \frac{1}{\sqrt{2}}(1+i)\beta \end{bmatrix}$$

## Action of Gates on Special States:

$$Z |+\rangle = |-\rangle, \quad Z |-\rangle = |+\rangle, \quad S |+\rangle = |i+\rangle$$

$$Z |i-\rangle = SS |i-\rangle = TTTT |i-\rangle = |i+\rangle$$

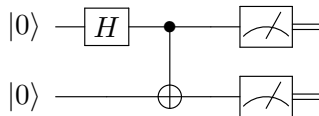
# Quantum Circuits

**Definition:** A quantum circuit is a computational model in which quantum gates are applied to qubits to perform quantum computations.

## Components of a Quantum Circuit:

- Qubits
- Quantum Gates (e.g.,  $H$ ,  $X$ ,  $Z$ ,  $CNOT$ ,  $T$ )
- Measurements (collapsing qubits into classical bits)

## Example: Bell State Circuit



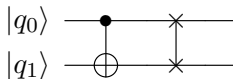
# Common Two-Qubit Quantum Gates

- CNOT (Controlled-NOT) Gate:

$$\text{CNOT} |11\rangle = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} |11\rangle = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = |10\rangle$$

- SWAP Gate:

$$\text{SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



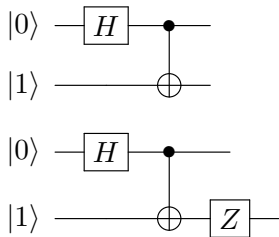
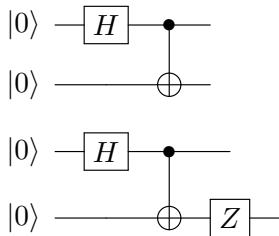
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# Bell States



$$CX|H|00\rangle = CX\left(\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|01\rangle\right) = \frac{1}{\sqrt{2}}CX|00\rangle + \frac{1}{\sqrt{2}}CX|01\rangle = \\ = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle = |\Phi^+\rangle$$

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|10\rangle + |01\rangle)$$

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|10\rangle - |01\rangle)$$

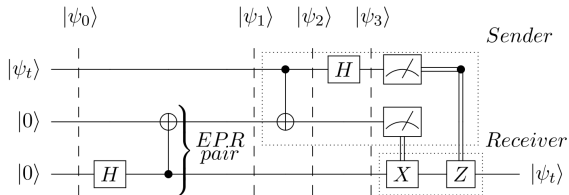
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# Quantum teleportation



$$|\psi_t\rangle = \alpha_t |0\rangle + \beta_t |1\rangle \quad |\psi_0\rangle = |\psi_t\rangle \otimes |00\rangle = \alpha_t |000\rangle + \beta_t |100\rangle$$

$$|\psi_1\rangle = \frac{\alpha_t}{\sqrt{2}}|000\rangle + \frac{\alpha_t}{\sqrt{2}}|011\rangle + \frac{\beta_t}{\sqrt{2}}|100\rangle + \frac{\beta_t}{\sqrt{2}}|111\rangle$$

$$|\psi_2\rangle = \frac{\alpha_t}{\sqrt{2}}|000\rangle + \frac{\alpha_t}{\sqrt{2}}|011\rangle + \frac{\beta_t}{\sqrt{2}}|110\rangle + \frac{\beta_t}{\sqrt{2}}|101\rangle$$

$$\begin{aligned} |\psi_3\rangle &= \frac{1}{2} |00\rangle \otimes (\alpha_t |0\rangle + \beta_t |1\rangle) + \frac{1}{2} |01\rangle \otimes (\alpha_t |1\rangle + \beta_t |0\rangle) + \\ &\quad + \frac{1}{2} |10\rangle \otimes (\alpha_t |0\rangle - \beta_t |1\rangle) + \frac{1}{2} |11\rangle \otimes (\alpha_t |1\rangle - \beta_t |0\rangle) = \\ &= \frac{1}{2} |00\rangle \otimes |\psi_t\rangle + \frac{1}{2} |01\rangle \otimes |\overline{\psi_t}\rangle + \frac{1}{2} |10\rangle \otimes |\psi_t^\dagger\rangle + \frac{1}{2} |11\rangle \otimes |\overline{\psi_t^\dagger}\rangle \end{aligned}$$

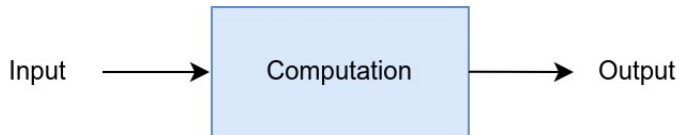
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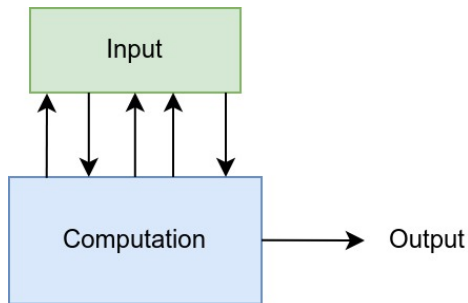
# Usual calculations



## Keypoint

The entire input is provided to the computation — nothing being hidden from the computation.

# The query model of computation



Input is in a form of function, which is accessed from computation via **queries**.  
The input can be provided by:

- **Oracle:** who knows everything, but answers only questions
- **Blackbox:** we don't know the analytical expression of function, but can evaluate arguments

# The query model of computation

## Functions:

$$f : \Sigma^n \rightarrow \Sigma^m$$

where  $n$  and  $m$  are positive integers and  $\Sigma = \{0, 1\}$

## Queries

The computation does a query when the function is evaluated once and  $f(x)$  can be used in the following calculations.

Efficiency is measured by counting the number of queries.

# Examples of query problems

Or

**Input:**  $f : \Sigma^n \rightarrow \Sigma$

**Output:** 1 if there exists a string  $x \in \Sigma^n$  for which  $f(x) = 1$ ,  
0 if there is no such string.

Parity

**Input:**  $f : \Sigma^n \rightarrow \Sigma$

**Output:** 0 if  $f(x) = 1$  for an even number of strings  $x \in \Sigma^n$ ,  
1 if  $f(x) = 1$  for an odd number of strings  $x \in \Sigma^n$ .

Minimum

**Input:**  $f : \Sigma^n \rightarrow \Sigma^m$

**Output:** The string  $y \in \{f(x) : x \in \Sigma^n\}$  that comes first in the  
lexicographic ordering of  $\Sigma^m$ .

# Examples of query problems

Sometimes we also consider query problems where we have a **promise** on the input. Inputs that don't satisfy the promise are considered as “don't care” inputs.

## Unique Search

### Unique Search

**Input:**  $f : \Sigma^n \rightarrow \Sigma$

**Promise:** There is exactly one string  $z \in \Sigma^n$  for which  $f(z) = 1$ ,  
with  $f(x) = 0$  for all strings  $x \neq z$

**Output:** The string  $z$

We sometimes consider very complicated and highly contrived problems to look for extremes that reveal potential advantages of quantum computing.

# Query Gates

Unitary implementation the problems (e.g. OR, Parity check ...), which allows the usage with quantum circuits.

## Definition

The query gate  $U_f$  for any function  $f : \Sigma^n \rightarrow \Sigma^m$  is defined as:

$$U_f(|y\rangle|x\rangle) = |y \oplus f(x)\rangle|x\rangle$$

for all  $x \in \Sigma^n$  and  $y \in \Sigma^m$ .

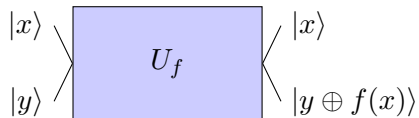
## Notation

The string  $y \oplus f(x)$  is the bitwise XOR of  $y$  and  $f(x)$ . For example:

$$001 \oplus 101 = 100$$

# Query Gates

In circuit diagrammatic form  $U_f$  operates like this:



This gate is always unitary, for any choice of the function  $f$ .

# Deutsch's Problem

Deutsch's problem is the **Parity** problem for functions of the form  $f : \Sigma \rightarrow \Sigma$ .

There are four functions of the form  $f : \Sigma \rightarrow \Sigma$ :

| $a$ | $f_1(a)$ | $a$ | $f_2(a)$ | $a$ | $f_3(a)$ | $a$ | $f_4(a)$ |
|-----|----------|-----|----------|-----|----------|-----|----------|
| 0   | 0        | 0   | 0        | 0   | 1        | 0   | 1        |
| 1   | 0        | 1   | 1        | 1   | 0        | 1   | 1        |

The functions  $f_1$  and  $f_4$  are **constant** while  $f_2$  and  $f_3$  are **balanced**.

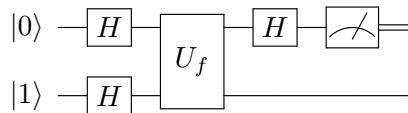
## Deutsch's Problem

**Input:**  $f : \Sigma \rightarrow \Sigma$

**Output:** 0 if  $f$  is constant, 1 if  $f$  is balanced.

# Deutsch's Algorithm

Deutsch's algorithm solves **Deutsch's problem** using a single query.



## Output Interpretation

- 0 if  $f$  is constant
- 1 if  $f$  is balanced

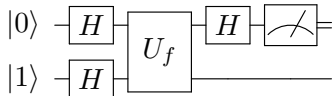
# Mathematical Representation

$$\begin{aligned} |\pi_1\rangle &= |-\rangle|+\rangle = \frac{1}{2}(|0\rangle - |1\rangle)|0\rangle + \frac{1}{2}(|0\rangle - |1\rangle)|1\rangle \\ |\pi_2\rangle &= \frac{1}{2}(|0 \oplus f(0)\rangle - |1 \oplus f(0)\rangle)|0\rangle + \frac{1}{2}(|0 \oplus f(1)\rangle - |1 \oplus f(1)\rangle)|1\rangle \\ &= \frac{1}{2}(-1)^{f(0)}(|0\rangle - |1\rangle)|0\rangle + \frac{1}{2}(-1)^{f(1)}(|0\rangle - |1\rangle)|1\rangle \\ &= |-\rangle \left( \frac{(-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle}{\sqrt{2}} \right) = \\ &= (-1)^{f(0)}|-\rangle \left( \frac{|0\rangle + (-1)^{f(0) \oplus f(1)}|1\rangle}{\sqrt{2}} \right) \\ &= \begin{cases} (-1)^{f(0)}|-\rangle|+\rangle & f(0) \oplus f(1) = 0 \\ (-1)^{f(0)}|-\rangle|-\rangle & f(0) \oplus f(1) = 1 \end{cases} \end{aligned}$$

# Mathematical Representation II

$$\begin{aligned} |\pi_2\rangle &= \begin{cases} (-1)^{f(0)}|-\rangle|+\rangle & f(0) \oplus f(1) = 0 \\ (-1)^{f(0)}|-\rangle|-\rangle & f(0) \oplus f(1) = 1 \end{cases} \\ |\pi_3\rangle &= \begin{cases} (-1)^{f(0)}|-\rangle|0\rangle & f(0) \oplus f(1) = 0 \\ (-1)^{f(0)}|-\rangle|1\rangle & f(0) \oplus f(1) = 1 \end{cases} \\ &= (-1)^{f(0)}|-\rangle|f(0) \oplus f(1)\rangle \end{aligned}$$

# Phase Kickback



## Mathematical Representation

$$|b \oplus c\rangle = X^c|b\rangle$$

$$U_f(|b\rangle|a\rangle) = |b \oplus f(a)\rangle|a\rangle = (X^{f(a)}|b\rangle)|a\rangle$$

$$U_f(|-\rangle|a\rangle) = (X^{f(a)}|-\rangle)|a\rangle = (-1)^{f(a)}|-\rangle|a\rangle$$

$$U_f(|-\rangle|a\rangle) = (-1)^{f(a)}|-\rangle|a\rangle \quad \leftarrow \text{phase kickback}$$

# The Deutsch-Jozsa Problem

The Deutsch-Jozsa problem generalizes Deutsch's problem: for an input function  $f : \Sigma^n \rightarrow \Sigma$ , the task is to output 0 if  $f$  is constant and 1 if  $f$  is balanced.

When  $n \geq 2$ , some functions  $f : \Sigma^n \rightarrow \Sigma$  are neither constant nor balanced.

## Example

This function is neither constant nor balanced:

| $\mathbf{x}$ | $\mathbf{f(x)}$ |
|--------------|-----------------|
| 00           | 0               |
| 01           | 0               |
| 10           | 0               |
| 11           | 1               |

Input functions that are neither constant nor balanced are “don't care” inputs.

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## Deutsch-Jozsa problem

**Input:**  $f : \Sigma^n \rightarrow \Sigma$

**Promise:**  $f$  is either constant or balanced

**Output:** 0 if  $f$  is constant, 1 if  $f$  is balanced

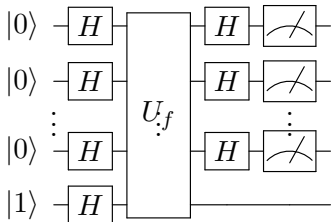
# The Deutsch-Jozsa Problem

## Deutsch-Jozsa problem

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# Deutsch-Jozsa Analysis

The Hadamard operation works like this on standard basis states:

$$H|0\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$H|1\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

We can express these two equations as one:

$$H|a\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}(-1)^a|1\rangle = \frac{1}{\sqrt{2}} \sum_{b \in \{0,1\}} (-1)^{ab} |b\rangle$$

# Deutsch-Jozsa Analysis

The Hadamard operation works like this on standard basis states:

$$H|a\rangle = \frac{1}{\sqrt{2}} \sum_{b \in \{0,1\}} (-1)^{ab} |b\rangle$$

Now suppose we perform a Hadamard operation on each of  $n$  qubits:

$$\begin{aligned} & H^{\otimes n} |x_{n-1} \cdots x_1 x_0\rangle \\ &= (H|x_{n-1}\rangle) \otimes \cdots \otimes (H|x_0\rangle) \\ &= \left( \frac{1}{\sqrt{2}} \sum_{y_{n-1} \in \Sigma} (-1)^{x_{n-1}y_{n-1}} |y_{n-1}\rangle \right) \otimes \cdots \otimes \left( \frac{1}{\sqrt{2}} \sum_{y_0 \in \Sigma} (-1)^{x_0y_0} |y_0\rangle \right) \\ &= \frac{1}{\sqrt{2^n}} \sum_{y_{n-1} \cdots y_0 \in \Sigma^n} (-1)^{x_{n-1}y_{n-1} + \cdots + x_0y_0} |y_{n-1} \cdots y_0\rangle \end{aligned}$$

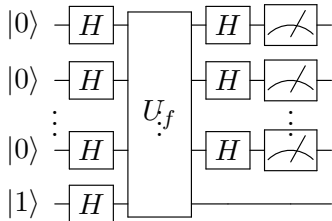
$$H^{\otimes n} |x_{n-1} \cdots x_1 x_0\rangle = \frac{1}{\sqrt{2^n}} \sum_{y \in \Sigma^n} (-1)^{x \cdot y} |y\rangle$$

For binary strings  $x = x_{n-1} \cdots x_0$  and  $y = y_{n-1} \cdots y_0$  we define:

$$\begin{aligned} x \cdot y &= x_{n-1}y_{n-1} \oplus \cdots \oplus x_0y_0 \\ &= \begin{cases} 1 & \text{if } x_{n-1}y_{n-1} + \cdots + x_0y_0 \text{ is odd} \\ 0 & \text{if } x_{n-1}y_{n-1} + \cdots + x_0y_0 \text{ is even} \end{cases} \end{aligned}$$

# Deutsch-Jozsa Analysis

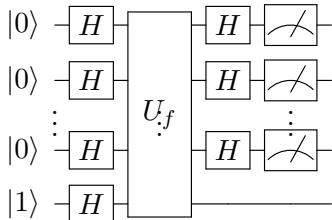
$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y \in \Sigma^n} (-1)^{x \cdot y} |y\rangle$$



$$|\pi_1\rangle = |-\rangle \otimes \frac{1}{\sqrt{2^n}} \sum_{x \in \Sigma^n} |x\rangle$$

# Deutsch-Jozsa Analysis

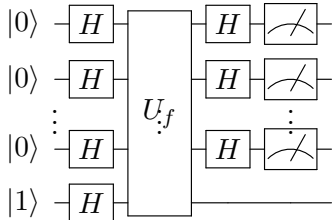
$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y \in \Sigma^n} (-1)^{x \cdot y} |y\rangle$$



$$|\pi_2\rangle = |-\rangle \otimes \frac{1}{\sqrt{2^n}} \sum_{x \in \Sigma^n} (-1)^{f(x)} |x\rangle$$

# Deutsch-Jozsa Analysis

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y \in \Sigma^n} (-1)^{x \cdot y} |y\rangle$$



$$|\pi_3\rangle = |-\rangle \otimes \frac{1}{2^n} \sum_{y \in \Sigma^n} \sum_{x \in \Sigma^n} (-1)^{f(x) + x \cdot y} |y\rangle$$

# Deutsch-Jozsa Analysis

The probability for the measurements to give  $y = 0^n$  is:

$$p(0^n) = \left| \frac{1}{2^n} \sum_{x \in \Sigma^n} (-1)^{f(x)} \right|^2 = \begin{cases} 1 & \text{if } f \text{ is constant} \\ 0 & \text{if } f \text{ is balanced} \end{cases}$$

The Deutsch-Jozsa algorithm therefore solves the Deutsch-Jozsa problem without error with a single query.

Any **deterministic** algorithm for the Deutsch-Jozsa problem must at least  $2^{n-1} + 1$  queries.

A **probabilistic** algorithm can, however, solve the Deutsch-Jozsa problem using just a few queries:

- 1 Choose  $k$  input strings  $x^1, \dots, x^k \in \Sigma^n$  uniformly at random.
- 2 If  $f(x^1) = \dots = f(x^k)$ , then answer 0 (constant), else answer 1 (balanced).

If  $f$  is constant, this algorithm is correct with probability 1.

If  $f$  is balanced, this algorithm is correct with probability  $1 - 2^{-k+1}$ .

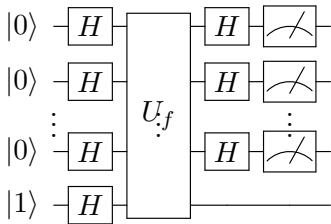
# The Bernstein-Vazirani Problem

## Bernstein-Vazirani problem

**Input:**  $f : \Sigma^n \rightarrow \Sigma$

**Promise:** there exists a binary string  $s = s_{n-1} \cdots s_0$  for which  
 $f(x) = s \cdot x$  for all  $x \in \Sigma^n$

**Output:** the string  $s$



# The Bernstein-Vazirani Problem

$$\begin{aligned} |\pi_3\rangle &= |-\rangle \otimes \frac{1}{2^n} \sum_{y \in \Sigma^n} \sum_{x \in \Sigma^n} (-1)^{f(x)+x \cdot y} |y\rangle \\ &= |-\rangle \otimes \frac{1}{2^n} \sum_{y \in \Sigma^n} \sum_{x \in \Sigma^n} (-1)^{s \cdot x + y \cdot x} |y\rangle \\ &= |-\rangle \otimes \frac{1}{2^n} \sum_{y \in \Sigma^n} \sum_{x \in \Sigma^n} (-1)^{(s \oplus y) \cdot x} |y\rangle \\ &= |-\rangle \otimes |s\rangle \end{aligned}$$

The Deutsch-Jozsa circuit therefore solves the Bernstein-Vazirani problem with a single query.

Any probabilistic algorithm must make at least  $n$  queries to find  $s$ .

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- Introduction to Quantum Computers and Quantum Computing
- Quantum entanglement
- Quantum teleportation
- Bernstein-Vazirani algorithm
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- Q&A and Closing the day

# Simon's Problem

## Simon's problem

**Input:** A function  $f : \Sigma^n \rightarrow \Sigma^m$

**Promise:** There exists a string  $s \in \Sigma^n$  such that

$$f(x) = f(y) \iff (x = y) \text{ or } (x \oplus s = y)$$

for all  $x, y \in \Sigma^n$

**Output:** The string  $s$

**Case 1:**  $s = 0^n$

The condition in the promise simplifies to:

$$f(x) = f(y) \iff x = y$$

This is equivalent to  $f$  being *one-to-one*.

# Simon's Problem

## Case 2: $s \neq 0^n$

The function  $f$  must be *two-to-one* to satisfy the promise:

$$f(x) = f(x \oplus s)$$

with *distinct output strings* for each pair.

| $x$ | $f(x)$ |
|-----|--------|
| 000 | 10011  |
| 001 | 00101  |
| 010 | 00101  |
| 011 | 10011  |
| 100 | 11010  |
| 101 | 00001  |
| 110 | 00001  |
| 111 | 11010  |

$$s = 011$$

$$f(000) = f(011) = 10011$$

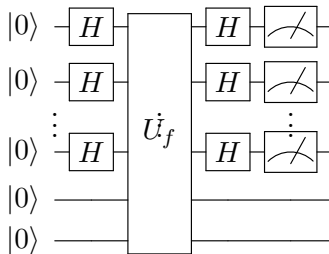
$$f(001) = f(010) = 00101$$

$$f(100) = f(111) = 11010$$

$$f(101) = f(110) = 00001$$

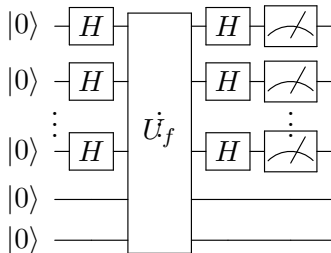
# Simon's Algorithm

Simon's algorithm consists of running the following circuit several times, followed by a post-processing step.



Measurements yield results  $y \in \Sigma^n$ .

# Simon's Algorithm Analysis



Measurements yield results  $y \in \Sigma^n$ .

$$|\pi_1\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \Sigma^n} |0^m\rangle |x\rangle \quad |\pi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \Sigma^n} |f(x)\rangle |x\rangle$$

$$|\pi_3\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \Sigma^n} |f(x)\rangle \otimes \left( \frac{1}{\sqrt{2^n}} \sum_{y \in \Sigma^n} (-1)^{x \cdot y} |y\rangle \right) = \frac{1}{2^n} \sum_{y \in \Sigma^n} \sum_{x \in \Sigma^n} (-1)^{x \cdot y} |f(x)\rangle |y\rangle$$

# Simon's Algorithm Analysis

$$\frac{1}{2^n} \sum_{y \in \Sigma^n} \sum_{x \in \Sigma^n} (-1)^{x \cdot y} |f(x)\rangle |y\rangle$$

$$\begin{aligned} p(y) &= \left\| \frac{1}{2^n} \sum_{x \in \Sigma^n} (-1)^{x \cdot y} |f(x)\rangle \right\|^2 = \left\| \frac{1}{2^n} \sum_{z \in \text{range}(f)} \left( \sum_{x \in f^{-1}(z)} (-1)^{x \cdot y} \right) |z\rangle \right\|^2 = \\ &= \frac{1}{2^{2n}} \sum_{z \in \text{range}(f)} \left| \sum_{x \in f^{-1}(z)} (-1)^{x \cdot y} \right|^2 \end{aligned}$$

$$\text{range}(f) = \{f(x) \mid x \in \Sigma^n\}$$

$$f^{-1}(\{z\}) = \{x \in \Sigma^n \mid f(x) = z\}$$

# Simon's Algorithm Analysis

$$p(y) = \frac{1}{2^{2n}} \sum_{z \in \text{range}(f)} \left| \sum_{x \in f^{-1}(z)} (-1)^{x \cdot y} \right|^2$$

**Case 1:**  $s = 0^n$  Because  $f$  is one-to-one, there is a single element  $x \in f^{-1}(\{z\})$  for every  $z \in \text{range}(f)$ :

$$\left| \sum_{x \in f^{-1}(\{z\})} (-1)^{x \cdot y} \right|^2 = 1$$

There are  $2^n$  elements in  $\text{range}(f)$ , so

$$p(y) = \frac{1}{2^{2n}} \cdot 2^n = \frac{1}{2^n}, \text{ for every } y \in \Sigma^n$$

# Simon's Algorithm Analysis II

**Case 2:**  $s \neq 0^n$  There are two strings in the set  $f^{-1}(\{z\})$  for each  $z \in \text{range}(f)$ ; if  $w \in f^{-1}(\{z\})$  either one of them, then  $w \oplus s$  is the other.

$$\begin{aligned} \left| \sum_{x \in f^{-1}(\{z\})} (-1)^{x \cdot y} \right|^2 &= \left| (-1)^{w \cdot y} + (-1)^{(w \oplus s) \cdot y} \right|^2 \\ &= |1 + (-1)^{s \cdot y}|^2 = \begin{cases} 4 & s \cdot y = 0 \\ 0 & s \cdot y = 1 \end{cases} \end{aligned}$$

There are  $2^{n-1}$  elements in  $\text{range}(f)$ , so

$$p(y) = \frac{1}{2^{2n}} \sum_{z \in \text{range}(f)} \left| \sum_{x \in f^{-1}(z)} (-1)^{x \cdot y} \right|^2 = \begin{cases} \frac{1}{2^{n-1}} & s \cdot y = 0 \\ 0 & s \cdot y = 1 \end{cases}$$

# Classical Post-Processing

Running the circuit from Simon's algorithm one time gives us a random string  $y \in \Sigma^n$ .

**Case 1:**  $s = 0^n$

$$p(y) = \frac{1}{2^n}$$

**Case 2:**  $s \neq 0^n$

$$p(y) = \begin{cases} \frac{1}{2^{n-1}} & s \cdot y = 0 \\ 0 & s \cdot y = 1 \end{cases}$$

Suppose we run the circuit independently  $k = n + r$  times, obtaining strings  $y^1, \dots, y^k$ .

# Classical Post-Processing II

$$\mathbf{y}^1 = y_{n-1}^1 \cdots y_0^1$$

$$\mathbf{y}^2 = y_{n-1}^2 \cdots y_0^2$$

$$\vdots$$

$$\mathbf{y}^k = y_{n-1}^k \cdots y_0^k$$

$$\mathbf{M} = \begin{pmatrix} y_{n-1}^1 & \cdots & y_0^1 \\ y_{n-1}^2 & \cdots & y_0^2 \\ \vdots & \ddots & \vdots \\ y_{n-1}^k & \cdots & y_0^k \end{pmatrix}$$

$$\mathbf{M} \begin{pmatrix} s_{n-1} \\ \vdots \\ s_0 \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

Using *Gaussian elimination*, we can efficiently compute the *null space* (modulo 2) of  $M$ . With probability greater than  $1 - 2^{-r}$ , it will be  $\{0^n, s\}$ .

## Classical Difficulty

Any probabilistic algorithm making fewer than  $2^{n/2-1} - 1$  queries will fail to solve Simon's problem with probability at least  $1/2$ .

- Simon's algorithm solves Simon's problem with a *linear* number of queries.
- Every classical algorithm for Simon's problem requires an *exponential* number of queries.

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# Q&A and Closing the day